SMART-FACILITY INCIDENT DETECTION · ASSESSMENT PROTOTYPE

Factory Safety Sentinel

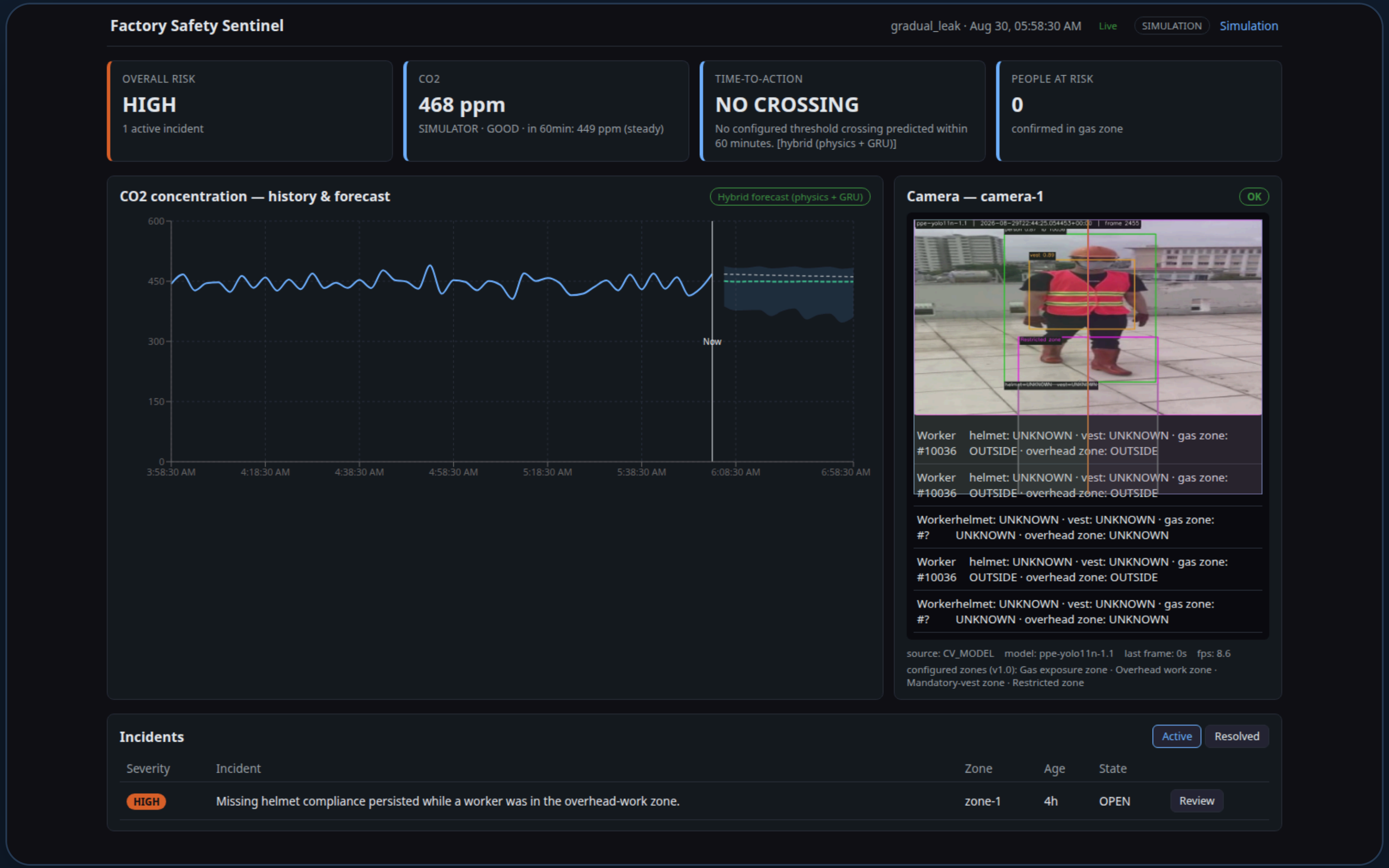
A predictive, multimodal industrial-safety system — physics- and ML-driven gas-risk forecasting fused with real-time computer-vision PPE and zone supervision, wired into one explainable incident and review workflow.

PHYSICS + XGBOOST + GRU · YOLO11N + BYTETRACK · FULL-STACK, TESTED, EVIDENCE-BACKED

Prepared for the Smart-Facility Incident Detection assessment · Contact: mohammad.2003.1m@gmail.com
Prototype — not certified for real industrial safety decisions.

Industrial safety monitoring is fragmented and reactive

- **Gas leaks develop gradually.** A slow CO_2 buildup can cross a dangerous threshold before any human is watching the reading.
- **PPE violations expose workers to preventable risk** — a missing helmet or vest is only caught if someone happens to look.
- **Restricted areas need continuous supervision** that a manager cannot realistically provide across a whole floor, every shift.
- **Sensors, cameras, and incident logs are usually separate systems** — nothing correlates a gas trend with a camera event with a reviewable record.



The real, working dashboard: one screen, live gas trend + forecast + camera + incidents — replacing the fragmented view above.

MVP scope vs. certified future deployment

DELIVERED IN THIS MVP

- Sensor-based risk monitoring (seeded, replayable CO_2 simulator)
- Predictive analysis (physics mass-balance + GRU residual + XGBoost leak classifier)
- Computer vision (YOLO11n + ByteTrack PPE/zone supervision)
- Working manager dashboard with live gas/camera/incident views
- Fully functioning backend (FastAPI, SQLite, WebSocket, REST)
- Simulation / digital twin for testing without a physical factory
- Evidence capture + full incident review/audit workflow

DEFERRED TO CERTIFIED DEPLOYMENT

- Real factory sensor/camera hardware integration
- Multi-camera, multi-zone, multi-worker at production scale
- Regulatory certification for real safety-critical decisions
- Domain-adapted vision model trained on real factory footage
- Production security hardening (auth, encrypted evidence storage)

End-to-end architecture — every stage, both evidence paths

SENSOR PATH

Simulator engine (seeded, deterministic)
↓
Ingestion service — POST /sensor-readings (idempotent)
↓
sensor_readings (SQLite)
↓
Contracts + validation (Pydantic v2)
↓
Physics forecast (mass-balance ODE)
↓
GRU residual model / physics-only fallback
↓
XGBoost leak classifier / rule fallback

VISION PATH

Camera / bundled replay video
↓
Vision worker — YOLO11n detection
↓
ByteTrack — anonymous worker tracking
↓
One-to-one PPE association (helmet/vest)
↓
Zone membership (point-in-polygon, dwell)
↓
vision_evidence (CV_MODEL / SIMULATION_GROUND_TRUTH)
↓
Evidence storage — real frame or schematic

↓
Deterministic risk / severity policy (versioned rules)
↓
Incident service — dedup, audit trail

incidents / audit_events (SQLite, Alembic)

WebSocket hub (+ REST poll fallback)

React dashboard — /dashboard

Three.js digital twin — /simulation

Three trained ML models, one analytical model, three deterministic components

COMPONENT	TYPE	ROLE
XGBoost leak classifier	TRAINED ML	Calibrated leak-probability estimate from sliding-window sensor features
Residual GRU forecast model	TRAINED ML	Learns the physics model's residual error to sharpen the 60-min CO_2 forecast
YOLO11n PPE detector	TRAINED CV MODEL	Detects person / helmet / vest / no_helmet on camera frames
Physics mass-balance model	ANALYTICAL, NOT ML	First-principles gas concentration ODE — the safe fallback when ML is unavailable
ByteTrack	TRACKING ALGORITHM	Associates detections into anonymous, session-local worker tracks — not trained
Restricted-zone polygons	DETERMINISTIC GEOMETRY	Configured point-in-polygon regions, not a learned "zone" class
Risk / severity policy	DETERMINISTIC LOGIC	Versioned rule table converting evidence into severity + reason codes

Verified against models/registry.json and CLAUDE.md's Frozen Models table

Proactive prediction, not threshold-only reaction

Gas readings → Sliding history window → Physics forecast → GRU residual correction → Leak classifier → Deterministic risk level → Incident alert

Every stage runs on **every tick**, using the last 10 simulated hours of history. The forecast projects 60 minutes ahead — so a manager sees a predicted threshold crossing *before* it happens, with an explicit "Time-to-Action" estimate, not just a current-value alarm.

Time-to-Action: the equations behind the forecast

WELL-MIXED MASS BALANCE

$$dC/dt = Q/V \cdot (C_{in} - C) + G/V$$

STEADY-STATE CONCENTRATION

$$C_{ss} = C_{in} + G/Q$$

TIME CONSTANT

$$\tau = V/Q$$

CONCENTRATION RESPONSE

$$C(t) = C_{ss} + (C_0 - C_{ss}) \cdot e^{(-t/\tau)}$$

TIME-TO-ACTION

$$t = -\tau \cdot \ln((C_{thresh} - C_{ss})/(C_0 - C_{ss}))$$

WHAT THIS MEANS FOR A MANAGER

Given the zone's volume, ventilation rate, and current emission source, the system solves exactly when CO_2 will cross the NIOSH action reference — and reports it in plain minutes, e.g. *"CO_2 may reach the 5000 ppm action reference in 34 minutes."*

Deliberately called **Time-to-Action**, never "time to harm" — the 5000 ppm reference is an occupational action threshold, not an immediate-harm line (CLAUDE.md).

Forecast accuracy and leak-classifier quality

MATCHED PHYSICS-VS-HYBRID BENCHMARK (N=1092)

57.19ppm
PHYSICS-ONLY MAE

47.59ppm
HYBRID PHYSICS+GRU MAE

▼ 16.8% ERROR REDUCTION

Identical held-out scenario/point set for both — a fair, matched comparison (models/evaluation/gru_benchmark_report.json).

LEAK CLASSIFIER (XGBOOST, CALIBRATED)

0.965
PR-AUC

0.022
BRIER (CALIBRATED)

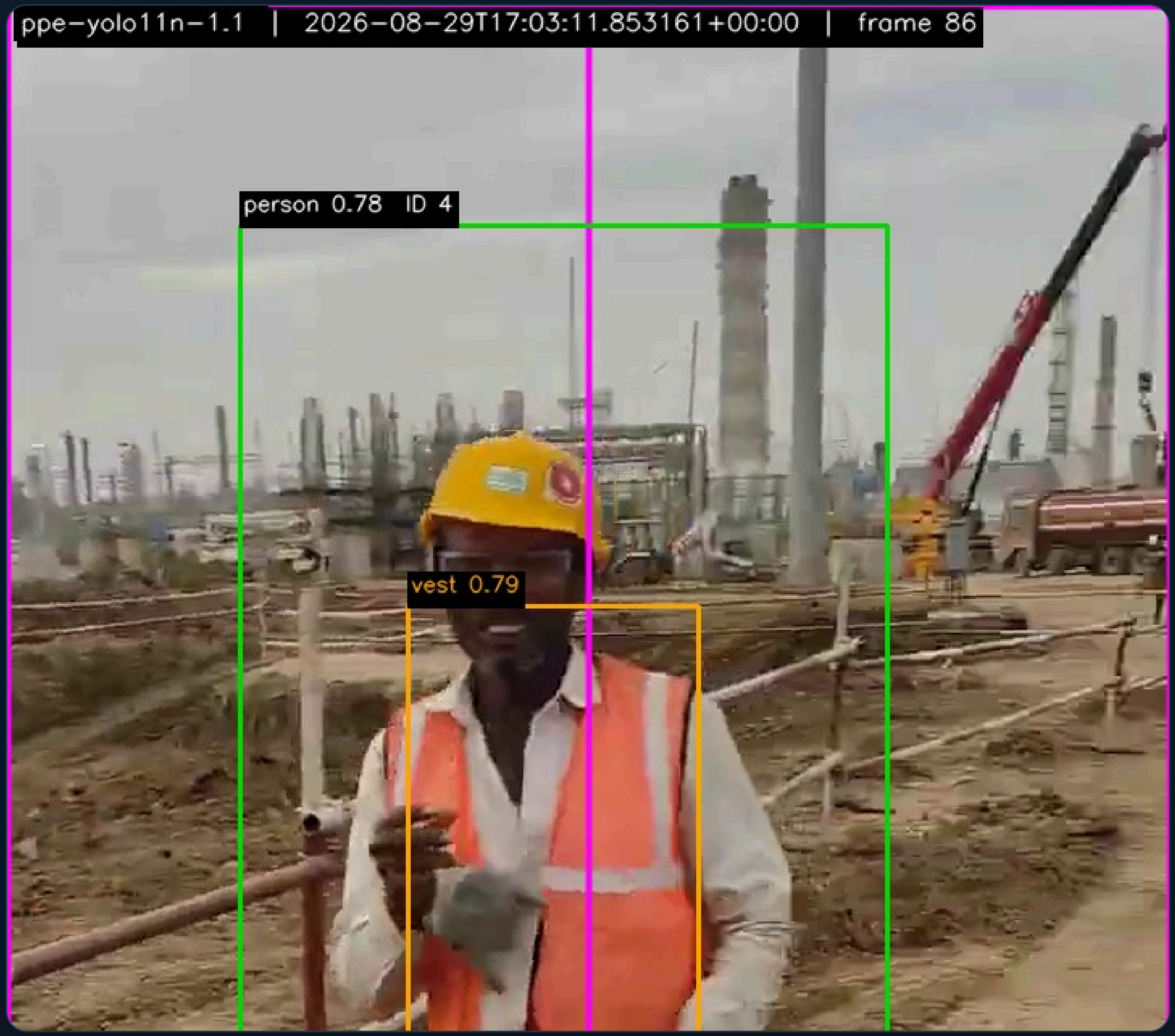
0.942
F1

n=900 held-out samples, 198 positive. Forecast inference latency: 1.79ms median / 2.20ms p95.

Not the same number: a separate, broader physics-only evaluation across 10 scenarios / 120 points reports 95.50 ppm MAE — a different, larger evaluation set. Never compared directly against the 47.59 ppm matched hybrid figure above.

YOLO11n + ByteTrack, real evidence, no fabricated CV

- **YOLO11n** — active version 1.1, 640px input, fine-tuned from COCO-pretrained weights
- **Class schema:** person · helmet · vest · no_helmet — the only 4 classes exposed at runtime
- **ByteTrack** assigns anonymous, session-local track IDs — no facial recognition, ever
- **PPE dwell:** 3s persistence required before a violation, 5s clear before resolving
- **Restricted-zone foot-point logic:** bottom-center of the person box tested against a configured polygon, with a 2s enter / 2s exit debounce
- **Real evidence capture:** the actual annotated camera frame that triggered the incident is saved — never a placeholder



Genuine annotated frame from the final interview-demo video: real person/PPE boxes, confidence scores, zone overlay, and burned-in model version + timestamp.

v1.1 active vs. v1.2 candidate — an honest promotion decision

WHY MORE DATA WAS TRIED

v1.1's `no_helmet` recall (0.175) was the known weak point. A second, independent dataset — **Industrial Safety** (Roboflow Universe, MIT license, 28,119 images) — was acquired to fine-tune a candidate.

Canonical mapping: hardhat → helmet, no_hardhat → no_helmet, person → person, safety_vest → vest. Deduplicated (3 exact dupes dropped) and leakage-checked before a class-balanced 8,000-image training subset was selected — augmented/near-duplicate video frames were capped, never counted as unique scenes.

PROMOTION DECISION: REJECTED

The candidate completed only 7 of a planned 50 epochs (externally interrupted for runtime, not early-stopped) and was evaluated as-is, on principle, without further training.

	V1.1 (ACTIVE)	V1.2 CANDIDATE
no_helmet recall	0.45	0.25
person recall	0.81	0.00

Gate failed on 2 of 3 checks — v1.1 remains active. Nothing overwritten silently.

Active model (v1.1) — held-out Construction-PPE test split

CLASS	PRECISION	RECALL	AP50	AP50-95
person	0.78	0.81	0.83	0.51
helmet	0.92	0.90	0.94	0.53
vest	0.84	0.87	0.89	0.57
no_helmet	0.38	0.17	0.23	0.09

0.562
MAP50

0.280
MAP50-95

LATENCY & VIDEO-EVENT PERFORMANCE

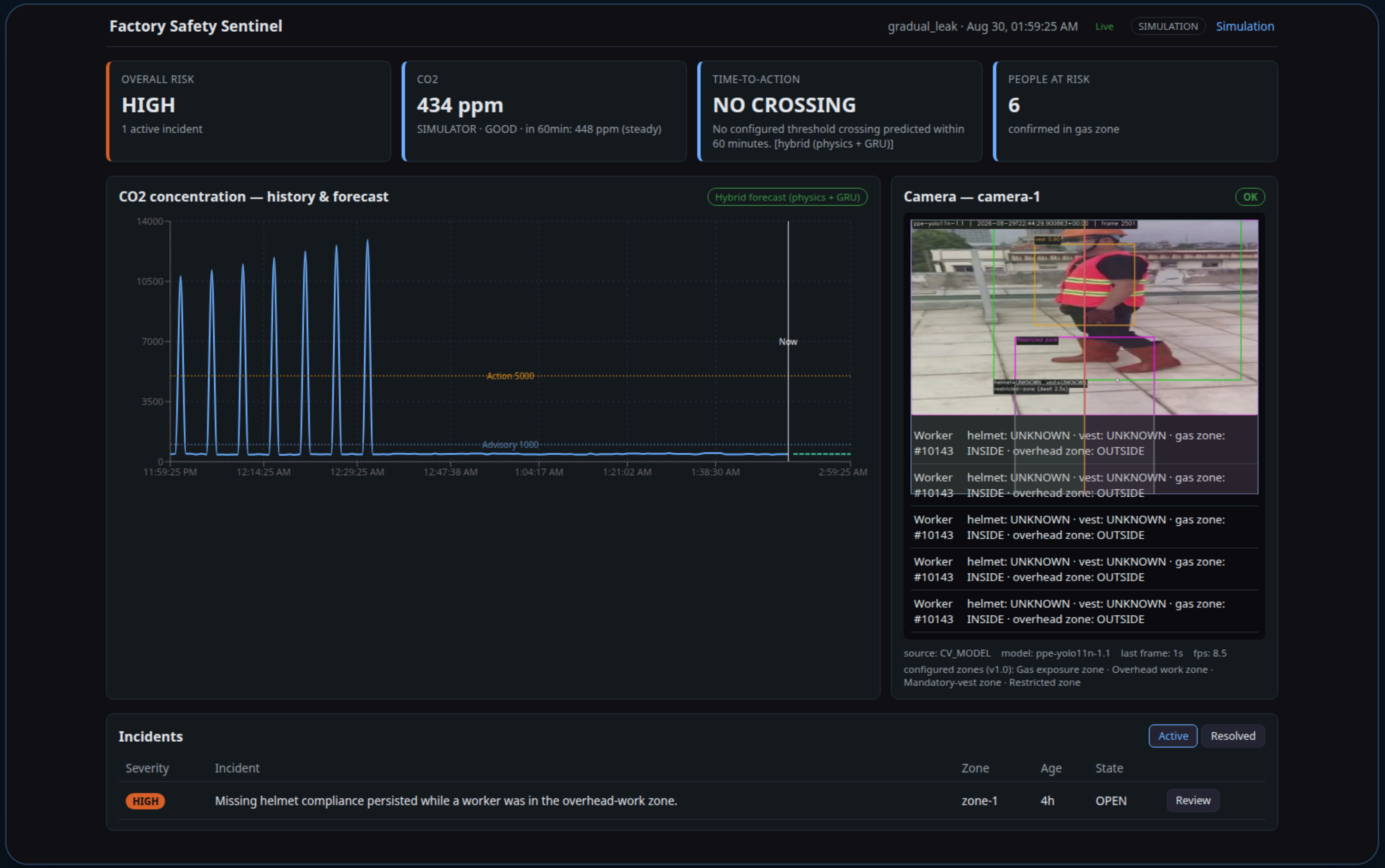
10.8ms
MEDIAN LATENCY (MX450)

90
ACHIEVED FPS

REAL-WORLD INTERVIEW-VIDEO STRESS TEST

Person detected in 354/527 frames, helmet in 143, vest in 401. **no_helmet fired in 0 frames** — a genuine, disclosed weakness (peaks at 0.049 vs. the 0.05 runtime threshold on this real clip).

One screen: live risk, forecast, camera, incidents



- **Live gas levels** — current CO2 ppm from real ingested readings
- **Risk severity** — deterministic overall status card (NORMAL / HIGH / CRITICAL…)
- **60-min forecast** — physics + GRU hybrid, with uncertainty band
- **Time-to-Action** — plain-language threshold-crossing estimate
- **Camera supervision** — live real detections, PPE/zone state per track
- **Active incidents** — severity, zone, age, one-click Review
- **Evidence thumbnails** — in the review drawer, real captured frames

Three genuine, real-video-triggered events

PPE COMPLIANT

Helmet + vest detected, confidence 0.78/0.79 — no incident.

PPE_HELMET_OVERHEAD_VIOLATION · HIGH

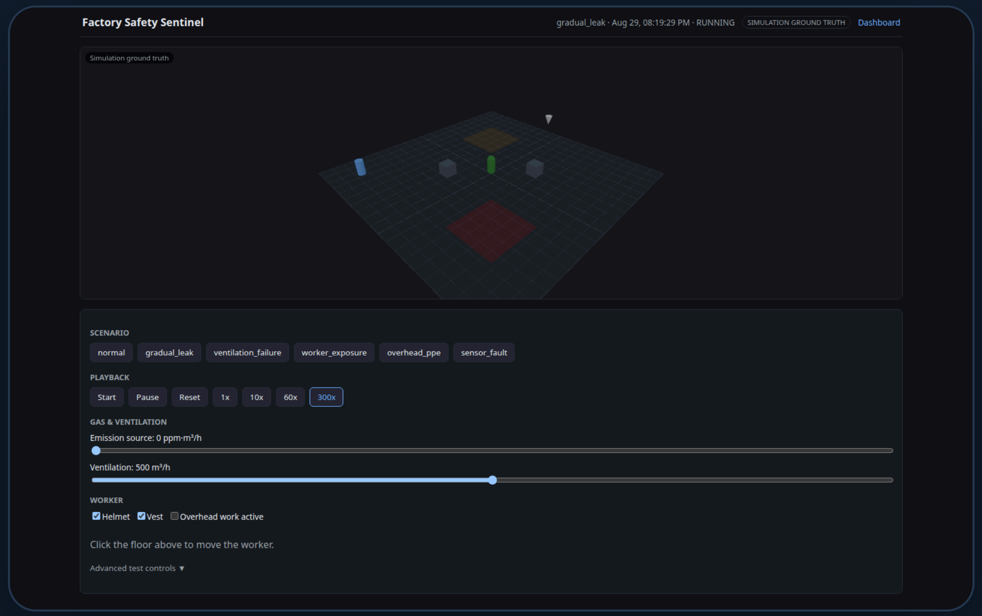
No positive helmet evidence while inside the overhead-work zone.

PERSON_IN_RESTRICTED_ZONE · HIGH

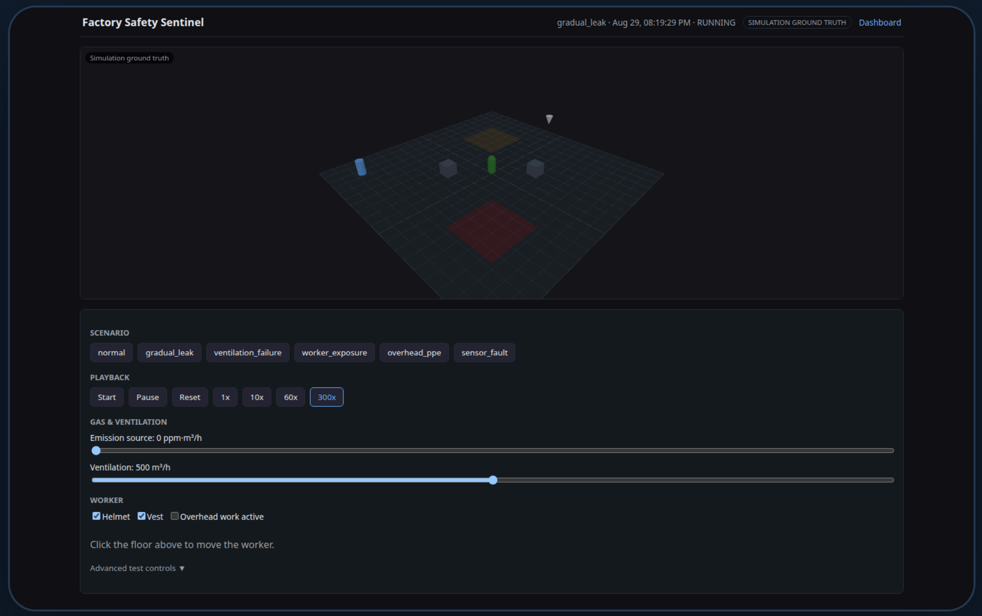
Foot point dwelled 2.5s inside the configured restricted polygon.

All three are real captured frames (`is_real_camera_frame=true`) from actual incidents opened by the live backend during this session — reviewable end-to-end with JSON/CSV reports in `docs/screenshots/final/07` and `/08`.

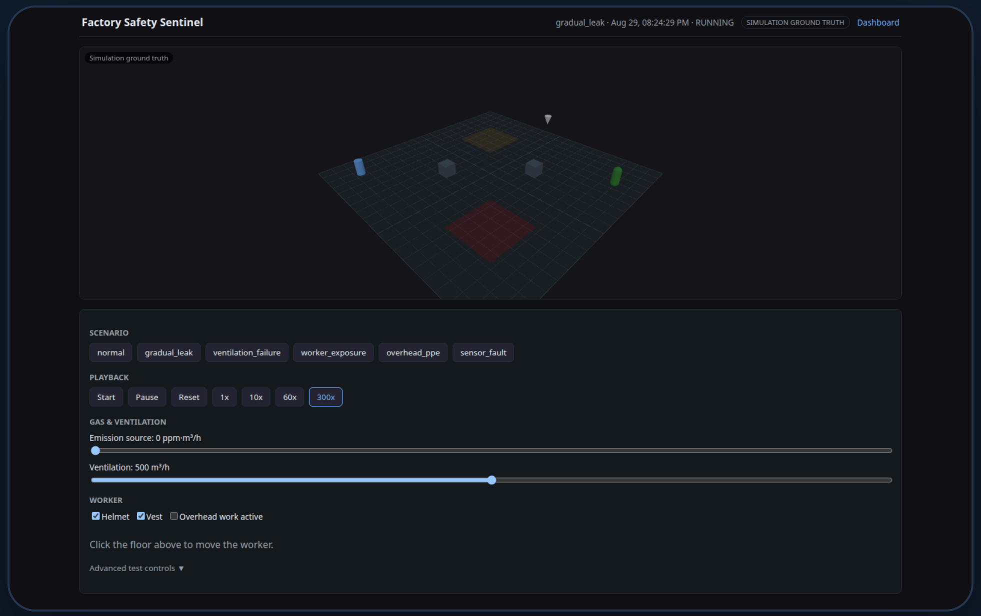
Testable without a physical factory



Factory layout: workers, zones, and machinery markers.



Live gas/ventilation sliders and scenario presets.



Click-to-move worker, helmet/vest toggles.

Simulation commands flow through **the exact same ingestion → risk → incident pipeline** a real sensor or camera would use — this is not a decorative mockup, it is a legitimate deterministic test harness (seeded, replayable, used by the automated test suite itself).

The real sequence, start to finish

1. Video / sensor input — real camera feed or seeded gas readings
2. Prediction / detection — YOLO+ByteTrack or physics+GRU+XGBoost
3. Deterministic risk decision — versioned severity policy
4. Incident creation — deduplicated, reason-coded, in the database
5. Evidence image — real captured frame or labelled schematic
6. Manager acknowledgement — Acknowledge → Investigate → Resolve
7. Downloadable report — JSON and CSV, per incident

WATCH IT RUN

Full interview-demo recording and annotated video are in the repository:

`github.com/M7mdknh/smart-detector`

deliverables/Factory_Safety_Sentinel_Interview_Demo.mp4

Verified this session, not stale figures

177
BACKEND TESTS PASSING

24
FRONTEND TESTS PASSING

2/2
PLAYWRIGHT E2E SUITES

0
KNOWN REGRESSIONS

ACCEPTANCE MATRIX (DOCS/ACCEPTANCE_RESULTS.MD)

A01–A18: 17 PASS, 1 PASS WITH LIMITATION, 0 FAIL · E01–E12: 10 PASS, 2 PASS WITH LIMITATION, 0 FAIL

Limitations are disk-quota constraints on a from-scratch vision install and evidence-image honesty labelling — never a functional failure. Docker build/up/down and clean-checkout `make demo` verified end to end in the same audit.

Stated directly, not defensively

- No real factory deployment data — trained/evaluated on construction-site imagery
- Construction-to-factory domain gap is real and measured, not assumed away
- No facial recognition — anonymous, session-local track IDs only, by design
- Evidence images contain worker imagery — retention/privacy policy needed for real use
- The model can and does miss PPE violations (no_helmet is a known weak point)
- **Not certified for real industrial safety decisions**
- A human manager remains responsible for every action taken
- Future work: calibration and domain adaptation on real factory footage

CONCLUSION

A proactive, evidence-based safety system — testable today, extendable tomorrow

- Proactive gas-risk prediction, minutes before a threshold is crossed
- Automated visual supervision of PPE and restricted zones
- Earlier response through a real, reviewable incident workflow
- Evidence-backed, auditable, never a fabricated alert
- Fully testable without a physical factory
- Extendable to real sensors and cameras with no backend rewrite

Questions?